



TobiasBelch

FOUNDING ENGINEER

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about

Founding engineer with 17 years shipping production platforms from concept to deployment in healthcare, e-health commerce, and sustainability. Builds from scratch — migrated a monolith to a multi-tenant platform serving 11M+ users, co-designed a domain model that unlocked 30 business cases, and established LLM agent patterns adopted across 3 teams. Carries deep full-stack capability (TypeScript, React, Node.js) with the versatility to own system design, infrastructure decisions, and cross-functional stakeholder communication. Effective in resource-constrained, high-ownership environments — equally comfortable at the keyboard, defining technical strategy, or aligning a small team around a shared goal.

skills

Full-Stack	Architecture & Infrastructure	AI, Quality & Leadership
TypeScript / JavaScript	System Design & Domain	LLM Integration / AI Agent
React / Next.js	Modelling	Orchestration
Node.js / NestJS	Frontend & BFF Architecture	Prompt Engineering
GraphQL / REST APIs	Multi-Tenant Platform Design	Design Systems
Redux / State Management	Docker / Git	Testing (Jest / Cypress /
Material UI / Tailwind / SCSS	AMQP / Apache Flink / NiFi	Playwright)
	IoT & Real-Time Systems	Technical Leadership &
		Mentoring
		D3 / Data Visualisation

languages

German (Native) · English (CEFR C-level)

education

work experience

Senior Software Engineer – DocMorris

Berlin, Germany | 08/2023 – Present

Digital healthcare and online pharmacy platform in the German market. ~11M active users.

- Led white-label monolith-to-multi-tenant migration — from architecture design through delivery. Onboarded 3 brands with ~90% code reuse; second tenant launched in 2 months vs. an estimated 2 years without the white-label architecture (~92% time reduction). Refactoring improved testability and maintainability over the original monolith.
- Established LLM agent patterns for product creation workflows from scratch — piloted within own team, then packaged and transferred to 3 additional teams. Improved response predictability and achieved an estimated 30x token-cost reduction on knowledge-retrieval tasks, with schema-validated outputs approaching deterministic reliability.
- Designed and led implementation of a cross-platform Design System (~30–40 components), structured from day one to enable AI-assisted UI generation. Projected to halve design time and reduce implementation effort by an estimated 80% through AI-powered component assembly.
- Drove frontend and BFF architecture evolution across the full stack — shaping technical direction, improving test coverage and accessibility compliance, and taking ownership of stakeholder communication in critical situations.
- Mentored 5 of 15 engineers through a structured program of 1:1s, pair programming, and weekly education sessions. One bootcamp-graduate promoted to mid-level within 2 years and to senior within 4.

Head of Engineering – Made2Flow

Berlin, Germany | 12/2022 – 07/2023

Sustainability platform for fashion supply-chain decision support. Startup environment (early-stage).

- Co-designed a next-generation domain data model with the CPO, replacing an MVP-stage model that blocked ~30 business cases. Delivered in 6 months alongside other responsibilities, directly unblocking several client-deals pending the transformation.
- Built and led a team of 4 engineers — hired 2 within 2 months in a competitive market. Established domain documentation and refinement processes that closed the gap between business goals and implementation, eliminating recurring friction between stakeholders and engineering.
- Owned technical partner communication and integration planning — managing relationships and delivery timelines across external partners.

Technical Lead – eHealth-Tec GmbH

Berlin, Germany | 04/2020 – 11/2022

Digital health products for patients, doctors, and pharmacists.

- Replaced ad-hoc architecture decisions — where proposals were lost and arbitrary shifts caused constant rework — with a structured ADR and design review process (~15–20 ADRs). Accelerated decision-making, improved codebase stability, reduced onboarding time for new engineers. Process still active 5 years on.
- Architected BFF layers insulating the frontend (~11M users) from restructuring across ~20 backend services. Eliminated ripple-effect changes when backend services were migrated or replaced, and provided the architectural seam for white-label tenant differentiation.
- Built and owned the frontend of a white-label online pharmacy platform serving 2 clients in production (TypeScript, React, Next.js).
- Led stakeholder communication and cross-functional alignment for a team of 4 in a Scrum-lite setup.
- Delivered a gematik-regulated browser-based e-prescription tool for physicians — working within strict regulatory constraints.

work experience

Head of Engineering – Medtower UG

Berlin, Germany | 09/2018 – 04/2020

Software and hardware solutions for hospital operations and visibility. Early-stage product.

- Architected a real-time distributed communication system using AMQP 1.0 / Apache Qpid where IoT hardware, servers, and web browsers operated as peers via pub/sub — eliminating point-to-point integration complexity. The protocol's finance-sector provenance ensured the uptime and reliability required in a hospital environment. This was the core infrastructure enabling all product use cases.
- Built 3D indoor multi-floor maps with live device location for hospital equipment tracking (beds, wheelchairs, devices), including indoor navigation and reporting views. Entire system operated in an offline hospital intranet — no internet access — requiring fully self-contained tile serving and real-time updates.
- Owned the full stack: microservices, SPA, live device tracking (Node.js, React, TypeScript), and IoT hardware integration.
- Managed an external Apache NiFi consultant for client-specific ETL pipelines — output enabled the team to build business cases within 2 weeks of engagement.
- Led a team of 2 engineers; established requirements engineering, refinement processes, and knowledge base from scratch.

earlier experience

Software Developer – remind me GmbH

02/2018 – 08/2018

Authored the technical concept for the white-label product transformation — adopted and shipped after tenure. Implemented feature-toggle architecture with lazy loading for flexible multi-context deployment.

Software Developer – Cedavis Technology GmbH

08/2009 – 01/2018

Over 9 years, sole owner of frontend architecture — introduced Angular 1, then migrated to React and TypeScript as the stack matured. Built a custom D3 chart component library from scratch. Engineered a PDF reporting module using Java Nashorn to render React components server-side, enabling a single shared codebase for live UI and PDF output. Implemented data simulation and forecasting features for healthcare capacity planning.

certifications

iSAQB Certified Professional for Software Architecture – Foundation

05/2022

oose

Domain Driven Design

12/2021

entwickler.de

